

NUCLEAR WARFARE

THE BOMBING OF HIROSHIMA

When US aircraft dropped atom bombs on the Japanese cities of Hiroshima and Nagasaki in August 1945 in a bid to end World War II, it opened a new perilous chapter in the history of warfare. For the first time, weapons had been created that were so destructive that their use had to be avoided at all costs.

The first test of a nuclear device took place at Alamogordo in the New Mexican desert on 16 July 1945. The explosion was equivalent to over 18,000 tonnes of TNT and generated temperatures three times hotter than the core of the Sun. The detonation was the culmination of the top secret Manhattan Project that had been initiated in December 1941 and headed by General Leslie Groves.

The objective of the project was to produce an atom bomb and deliver it on an enemy target, and to this end Groves had not only directed teams of scientists and engineers, but also set up a special flying group to drop the bomb. Group 509 of the US Army Air Force, commanded by Colonel Paul Tibbets, was equipped with the Boeing B-29 Superfortress, America's latest bomber aircraft, which had the range to strike Japanese cities from US-held Pacific island bases.

In late July and early August, components for a uranium-based atomic bomb, codenamed "Little Boy", and several plutonium-based bombs, codenamed "Fat Man", were delivered to the island of Tinian in the Marianas, where Colonel Tibbets' B-29s were based. The intention was to drop the bombs on selected Japanese cities as soon as the components had been assembled and when weather conditions permitted.

On 26 July, after the Potsdam conference attended by America, Britain, and the Soviet Union, the US and its allies called on Japan to surrender or face "prompt and utter destruction". When this call was rejected by Japan two days later, preparations for the world's first nuclear strike went ahead.

Hiroshima had been chosen as the first target as it was arguably of military significance, with a barracks and port, and was also largely undamaged, unlike most Japanese cities already devastated by conventional bombing. A large and flat urban area with a population of 300,000, Hiroshima would demonstrate the maximum effect of the bomb.

Colonel Tibbets chose to lead the operation in person. He piloted the B-29 designated to carry the "Little Boy" bomb and had his mother's maiden name, *Enola Gay*, written on the aircraft's side.

The *Enola Gay* was escorted by two other B-29s and arrived over Hiroshima at 8:15am on 6 August after an uneventful flight from Tinian. The aircraft released the bomb from an altitude of 30,000 feet, and it detonated in the air a minute later over the centre of the city.

TOTAL DESTRUCTION

The destruction surpassed all expectations: almost every building was destroyed over an area of 12 square km (7.5 miles) and estimates of the death toll range from 80,000 to 140,000. Of this number, those not killed immediately by the effect of the heat flash and blast died of radiation sickness over the following months.

Three days later the "Fat Man" plutonium bomb was dropped on another city, Nagasaki. It killed an estimated 35,000 to 80,000 people – a lower figure because hills had restricted the bomb's effectiveness. Japan surrendered on 15 August, heralding the end of World War II. No nuclear device has been used in warfare since.

